
GLOBAL WEALTH MANAGEMENT SYSTEMS: QUANTITATIVE TRADING DESK SPECIFICATIONS

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SECURITY CLASSIFICATION: INTERNAL USE ONLY

1. EXECUTIVE OBJECTIVE & SYSTEM CONSTRAINT BOUNDARIES

The primary purpose of the Multi-Asset Algorithmic Execution Terminal is to monitor, clean, and analyze high-frequency time-series datasets spanning four primary asset verticals: Equities, Benchmark Indices, Safe-Haven Commodities, and Forex Pairs. The core computational layer must maintain sub-second rendering speeds while executing over 56,000+ granular hourly transactional data records.

2. LOGISTICAL INGESTION AND PIPELINE BOUNDARIES

- Data Refresh Frequency: Standard daily rollup sheets are strictly prohibited. Ingestion pipelines must handshake with market APIs at fixed 1-Hour increments.
- Ingestion Range: Historical data tracking must be locked to a rolling window of exactly 729 trailing calendar days to optimize local storage caches and maintain data density without execution time lag.
- Fault Isolation: The ETL pipeline must incorporate automatic fallback schema-casting rules. Blank text rows (" ") or trailing null values generated during market-close intervals must be intercepted and force-cast via exception wrappers to zero values rather than letting queries crash the interface compilation.

3. FINANCIAL METRICS AND MATHEMATICAL VALIDATION LAWS

- Standalone Risk Profiling: Asset volatility indices must evaluate standard deviation across the full historical distribution, bypassing local date filters to retain a baseline risk footprint.
- Systematic Risk Evaluation (Beta Multipliers): Individual security covariance trends must be computed strictly relative to the market baseline anchor index (NSEI - Nifty 50). Ratios near 1.00 establish core index tracking behavior.
- Momentum Moving Averages Tracker: Trend-following moving average filters must run under 50-Hour and 200-Hour execution windows. The system must enforce dynamic harddatatype conversions to prevent string-to-number validation errors.
- Capitulation Metrics: Peak-to-trough drawdown parameters must execute real-time delta tracking on intra-day highest thresholds against hourly execution prices to flag immediate loss exposures.

4. USER INTERFACE GRAPHICAL ARCHITECTURE COMPLIANCE

- Spatial Density: The dashboard must feature a tightly packed layout framework. Grid systems must maintain strict left-to-right margin symmetries (X and Y position alignment) with a zero-gap policy for blank background spaces.
- Theme Contrast: Design presets must follow institutional fintech standards utilizing dark finishes (Midnight Blue #070B19, Card containers #10162D) combined with neon electric indicators (#00FFCC and #FF0055) to highlight critical threshold breaks.

- Inter-Page Navigation Grid: The workspace must deprecate native bottom page tabs. Switching must run under a top-aligned, synchronized horizontal navigation action button bar running page navigation hyperlink logic.

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END OF SPECIFICATION FILE - AUTHORIZED BY QUANT RISK MANAGEMENT DESK

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